

Miggy G. Rivera

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Education

Jose Rizal University

Bachelor of Science in Information Technology

2024 - PRESENT

Experience

AWS Learning Club JRU Ch : Web Development Lead

01/21/2026 - PRESENT

- Led a team of 10 developers in building the official AWS Learning Club JRU website using Vanilla JS, HTML5, CSS3, and Supabase, deployed on Vercel. Managed Git workflows, weekly sprints, and code reviews while mentoring members on Git, JS, and web fundamentals. Delivered a fully responsive, high-performance site (WebP, caching, pre-rendering) with an integrated AI chatbot, achieving fast load times and a polished UI/UX validated through user testing

AWS Learning Club JRU Ch : Vice President of External Affairs

04/01/2026 - PRESENT

- Leads the organization's external relations and partnership strategy, overseeing the full lifecycle from outreach to MOA execution. Onboarded 2 official partners (including AWS Learning Club – Polar Chapter), manages 5–10 partnership inquiries monthly, and represents the org at industry events (AWS, Canva, StellarX). Supervises the Relations Committee, drafts formal correspondence, and evaluates proposals against the org's CBL and community interests—focused on building long-term, mutually beneficial partnerships*

Technical Expertise

- Data Engineering & Systems: Python (3.12+), ETL Pipeline Architecture, SQL (PostgreSQL/SQLite), Vector Databases (Supabase pgvector), LlamaParse, Multi-modal Data Fusion (Sentinel Hub)
- Full-Stack Development: TypeScript, Next.js 14, React, Tailwind CSS, FastAPI, Flask, RESTful APIs (CRUD, JWT Auth)
- DevOps & Cloud: Docker, CI/CD (GitHub Actions), Vercel/Railway, API Rate Limiting (Slowapi), Linux/Bash
- AI & Analytics: RAG, Semantic & Hybrid Search, Risk Scoring Algorithms, Google Gemini/Vertex AI

Projects

Rule VII — AI Architect | Python, FastAPI, Supabase (pgvector), LlamaParse

- Engineered a **RAG-based ETL pipeline** to ingest complex PDFs (NBCP, Fire Code, BP 344) using LlamaParse for high-fidelity text extraction.
- Architected a vector search system with sentence-transformers and pgvector for hybrid (**semantic + keyword**) retrieval of architectural legal citations.
- Designed a PostgreSQL schema managing user metadata, project files, and embedded document chunks for real-time AI critiquing.
- Implemented JWT authentication and API rate limiting (20 req/min) to ensure backend security and scalability.

HYDRA: Flood Control Transparency System | Python, Flask, SQLite, Sentinel Hub

- Developed a **5-stage automated data pipeline** (Scrape → Validate → Geocode → Satellite → Database) to audit public infrastructure projects for fraud.
- Built a rule-based validation engine flagging high-risk projects via contractor history, timeline discrepancies, and cost anomalies.
- Integrated multi-source data fusion by syncing BetterGov project data with Sentinel Hub satellite imagery for site verification and geographic mapping.
- Engineered a quantitative risk-scoring system triaging thousands of projects into critical, high, and low-risk tiers based on weighted indicators and COA-flagged contractors.